Atmosic SDK

Quick Start Guide

SUMMARY: This document provides instructions on how to install the Atmosic Software Development Kit (SDK) and use it with the Atmosic Evaluation Kit (EVK) based on ATM2/ATM3 and ATM33/e Wireless SoC Devices.

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1. Overview

This document provides instructions on how to install the Atmosic Software Development Kit (SDK) to support Evaluation Kits (EVK) based on the ATM2/ATM3/ATM33/ATM33e Wireless SoC Devices. For more information about using the EVK, please refer to the EVK User Guides.

For third party development tool environments (IDEs), refer to IAR Workbench User Guide, SEGGER Embedded Studio User Guide, or Keil MDK Configuration Guide for installation and configuration, and IDE Auxiliary Flash Programming Tool User Guide available on the Atmosic Support website. Please submit a support request for access credentials to the Atmosic Support website.

Related documents are listed in the Reference Documents section.

2. EVK Setup

2.1 ATM2/ATM3 EVK

There are multiple versions of the ATM2 or ATM3 EVKs based on the specific device and package configuration. For hardware setup and configuration instructions, consult the User Guide for the specific EVK being used. The latest version of EVK User Guides can be found on the Atmosic Support website.

2.2 ATM33/e EVK

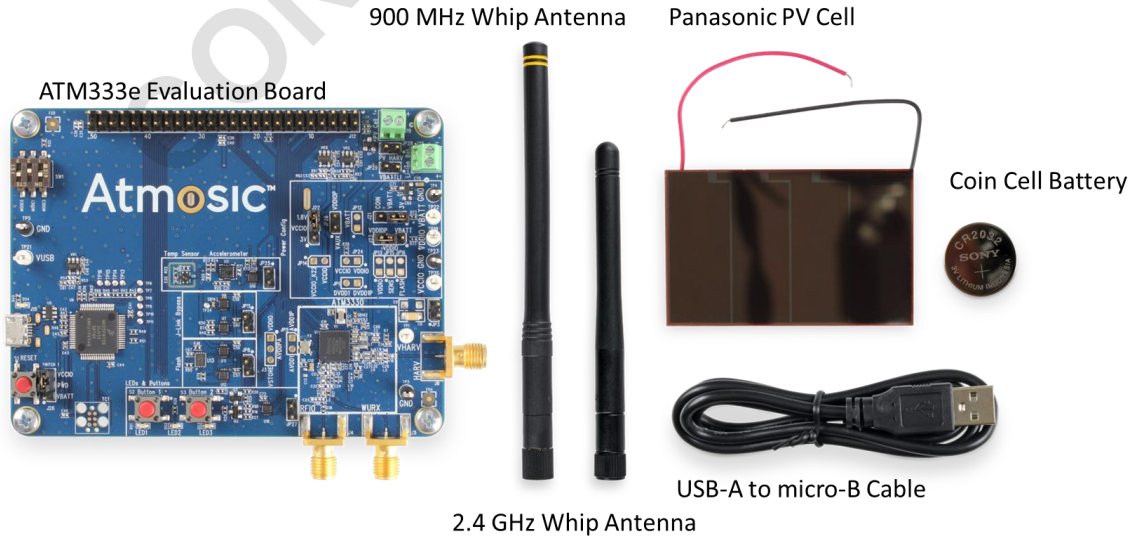


Figure 1 - ATM33/e Evaluation Kit

1) Connect the ATM33/e EVB Micro-B USB connector as shown in Figure 2 to the computer using the Micro-B USB cable provided.

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2) Attach the 2.4 GHz whip antenna to the RFIO connector as shown in Figure 3.

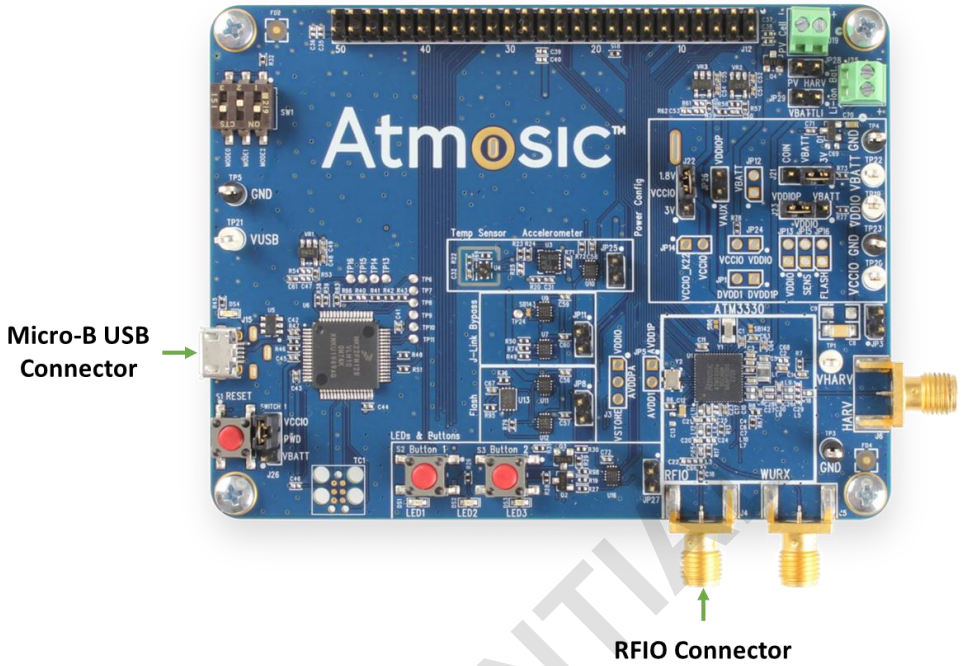


Figure 2 - ATM33/e Evaluation Board



Figure 3 - ATM33/e Evaluation Board with 2.4 GHz Whip Antenna

3) Install COIN battery when VBATT = COIN OR configure EVB to run on VBATT = 3 V. Refer to the ATM33/e Series Evaluation Kit User Guide for correct jumper settings.

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3. SDK Installation

3.1 Using Windows Installer

The SDK Windows Installer which supports Windows 10 and later is available on the Atmosic Support website. The installation requires administrator privileges.

3.2 Install Atmosic SDK

The setup package of Windows Installer for Atmosic SDK is a compressed file named AtmosicSDK_Inst_<version_number>.zip. Please extract the file and double click the AtmosicSDK_Inst_<version_number>.exe to start the Atmosic SDK Installer. Figure 4 shows the release package source tree after the files are extracted.

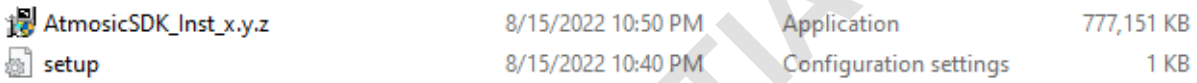


Figure 4 - Extracted Release Package

Please review the license agreement at Atmosic website. If you accept the terms of the agreement, click I Agree to continue.

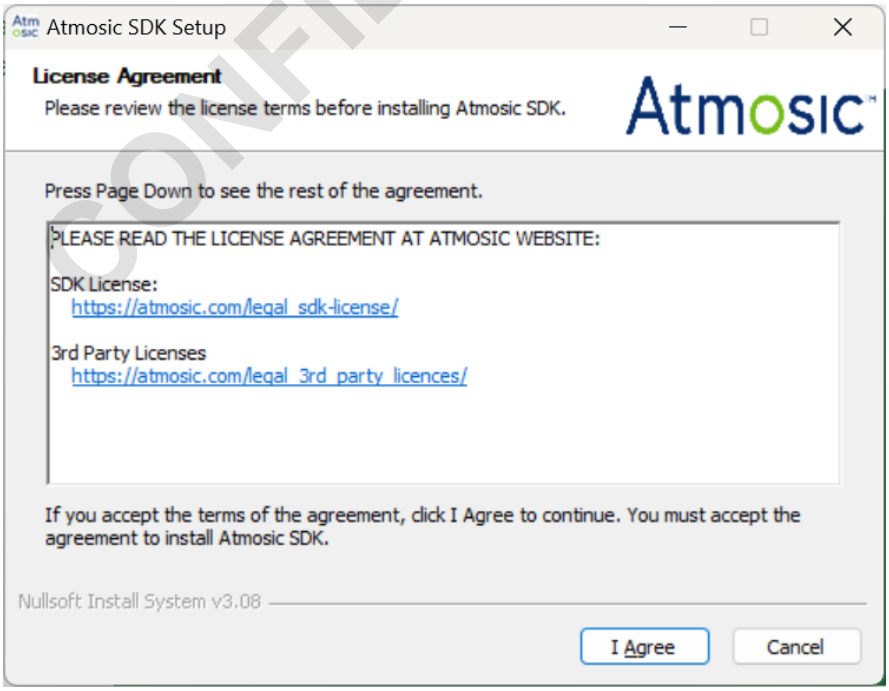


Figure 5 - Atmosic License Agreement

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The Atmosic SDK will occupy 2.5 GB disk space and will be located in the C:\AtmosicSDK\ folder, see Figure 6. Please make sure the disk space is enough before installing the Atmosic SDK. Subsequently, click the Install button to continue.

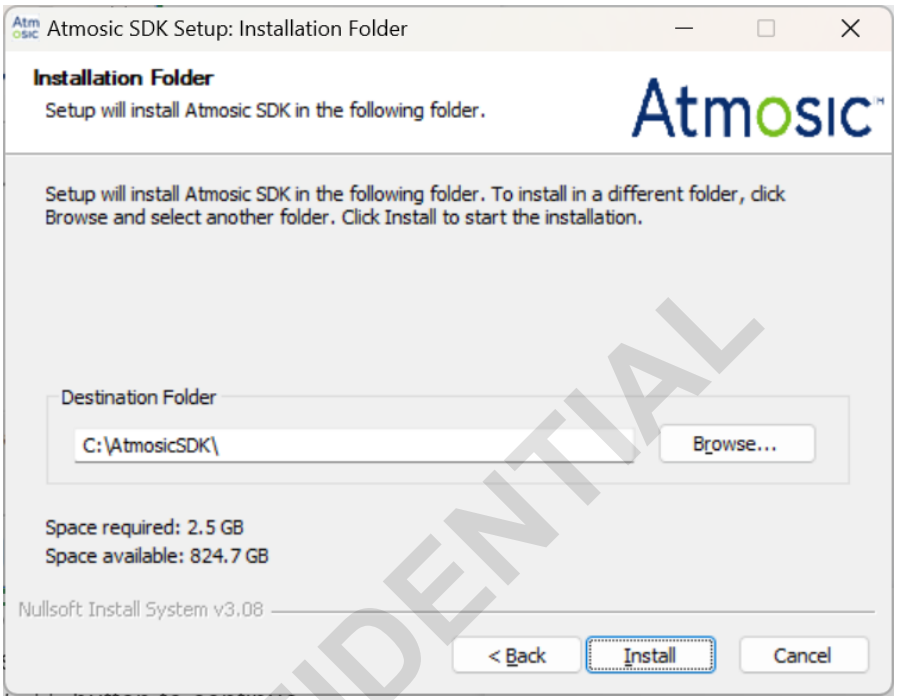


Figure 6 - SDK Installation Folder

***Note**: The installer cannot accept a path destination directory name with spaces. For example, Atmosic SDK is not acceptable while AtmosicSDK is acceptable. The Windows Installer will pop-up a message if the user selects an installation path name that includes spaces. See Figure 7.*

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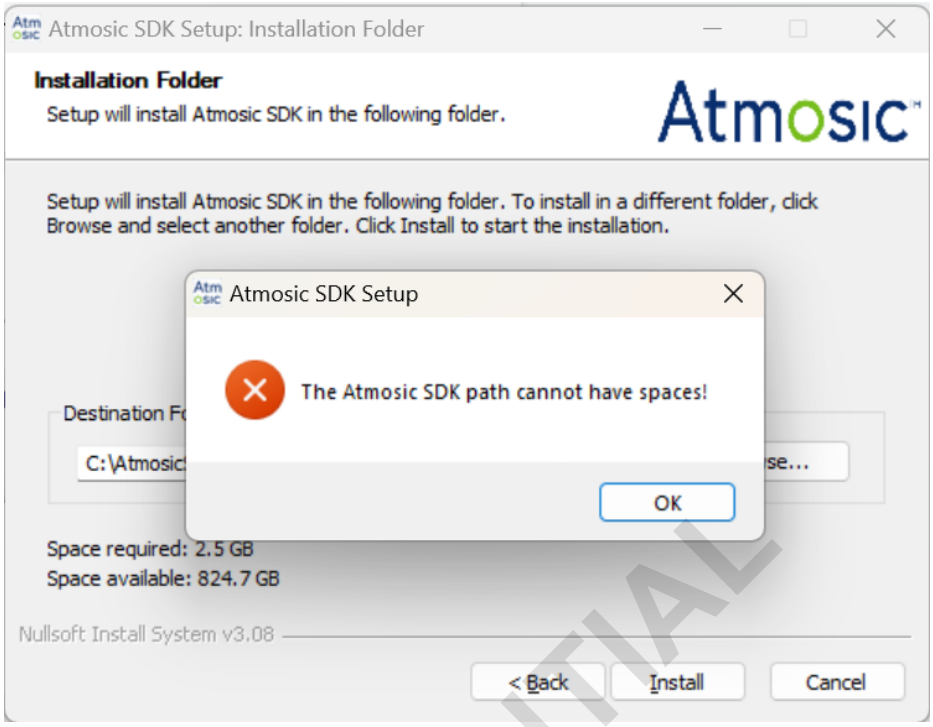


Figure 7 - SDK Installation Path Message

The installation process takes several minutes to complete. The Windows Installer is installing MSYS2, GNU Toolchain, and Atmosic SDK files to the target directory. See Figure 8.

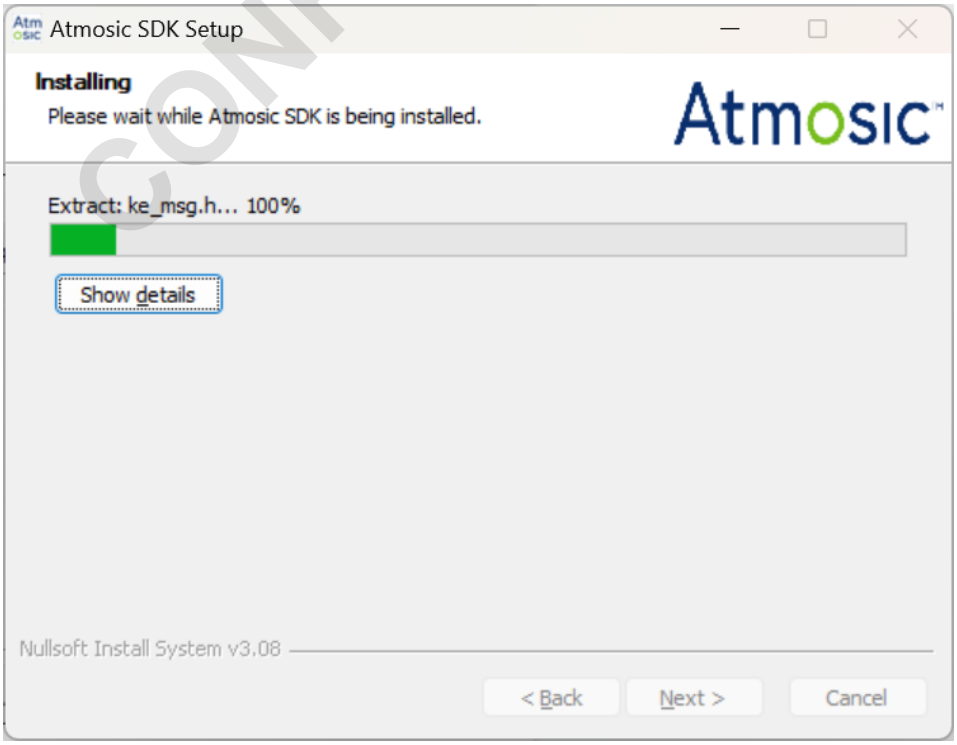


Figure 8 - Installing SDK

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The Windows Installer will pop up a message to notify the installation completed, click OK to close the Windows Installer. See Figure 9.

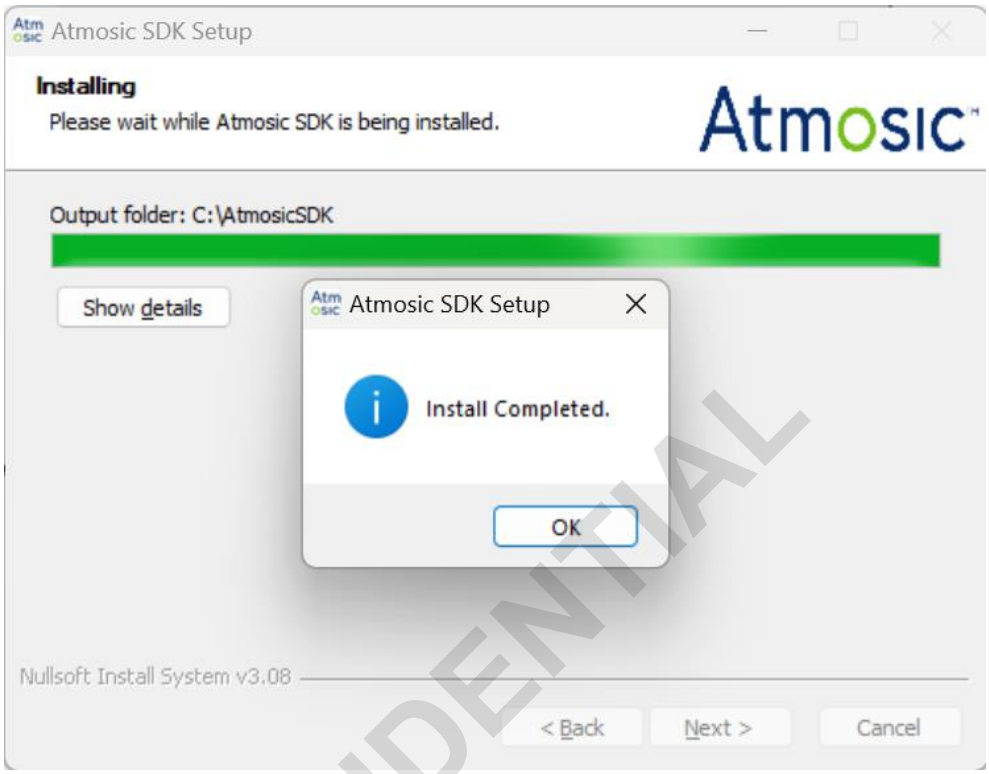


Figure 9 - Installation Completed

The Installer will create a folder named AtmosicSDK in the Start Menu as shown in Figure 10 during the installation process.

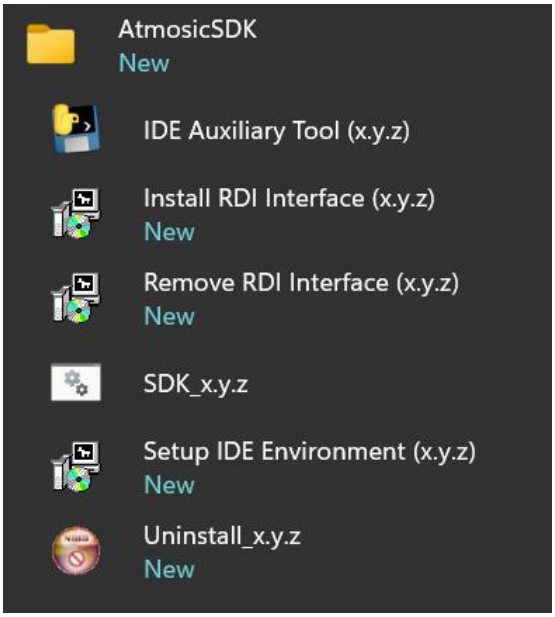


Figure 10 - SDK in Start Menu

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3.3 Install Atmosic Reference Design Interface (RDI)

3.3.1 ATM2/ATM3 EVK

Please connect the Atmosic EVB (USB1) to the Interface board, and connect the USB cable to the laptop and wait for Windows to install the FTDI driver (~ 30 seconds).

Open the device manager (open the Run dialog box by pressing and holding the Windows key, then press the R key).

Enter `devmgmt.msc` then click the OK button to verify whether the RDI USB interface exists or not.

If there are 2 COM ports listed in the device manager after plugging-in the Atmosic EVB to the laptop, please click the Install RDI Interface shortcut in `Start Menu/Atmosic` folder. See Figure 11.

***Note**: The COM port numbers are assigned by the Windows system.*

***Note**: If the device manager shows Atmosic RDI USB driver is already installed, move to the Connect To Debug Interface section.*

***Note**: Atmosic RDI USB driver installation is required when the new interface board is connected to the computer for the first time.*

***Note**: Refer to the Reference Documents section for more information regarding the Atmosic EVKs.*

The Device Manager shows COM port information before installing the Atmosic RDI USB driver.

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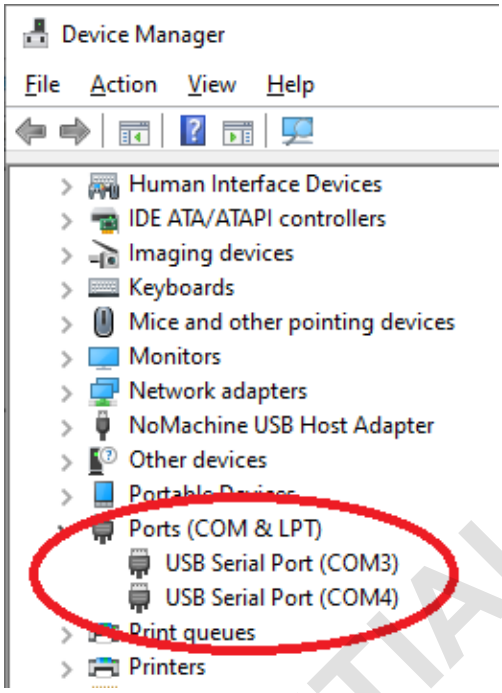


Figure 11 - USB COM Port with ATM2/ATM3 EVK

Please click the `Install` button if the system pop-up the following message.

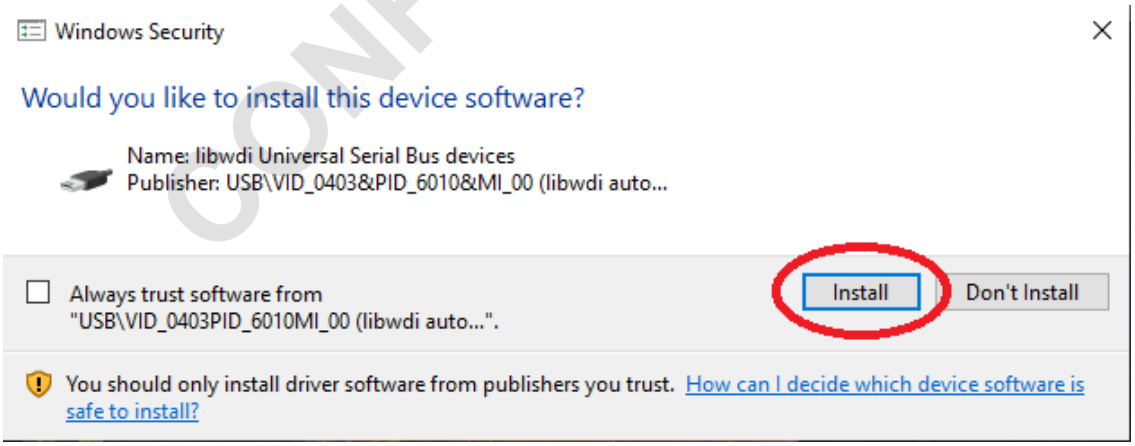


Figure 12 - Install RDI USB Driver with ATM2/ATM3 EVK

After installing Atmosic RDI USB driver, the Atmosic RDI USB1 interface will display in the Device Manager. Moreover, the COM4 is a debug console that could be used to display debug messages through PuTTY.

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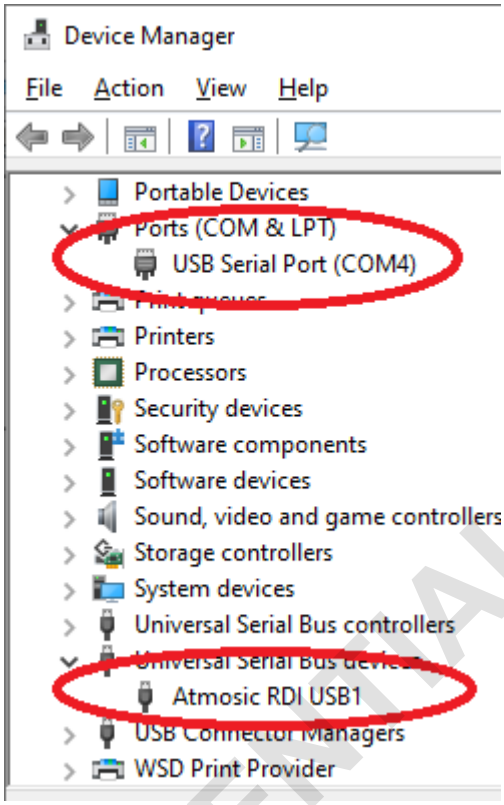


Figure 13 - RDI USB in Device Manager with ATM2/ATM3 EVK

3.3.2 ATM33/e EVK

Please connect the Atmosic EVB to the laptop and wait ~ 30 seconds. First time installation, a J-Link notification will pop-up in Figure 14. Allow it to finish setting up as shown in Figure 14. This will not be displayed in the device manager.

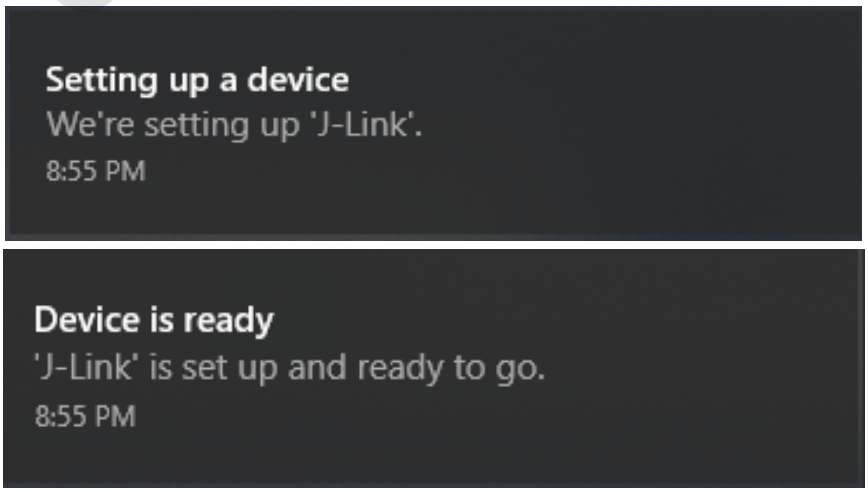


Figure 14 - J-Link Notifications with ATM33/e EVK

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Open the device manager (open the `Run` dialog box by pressing and holding the Windows key, then press the `R` key).

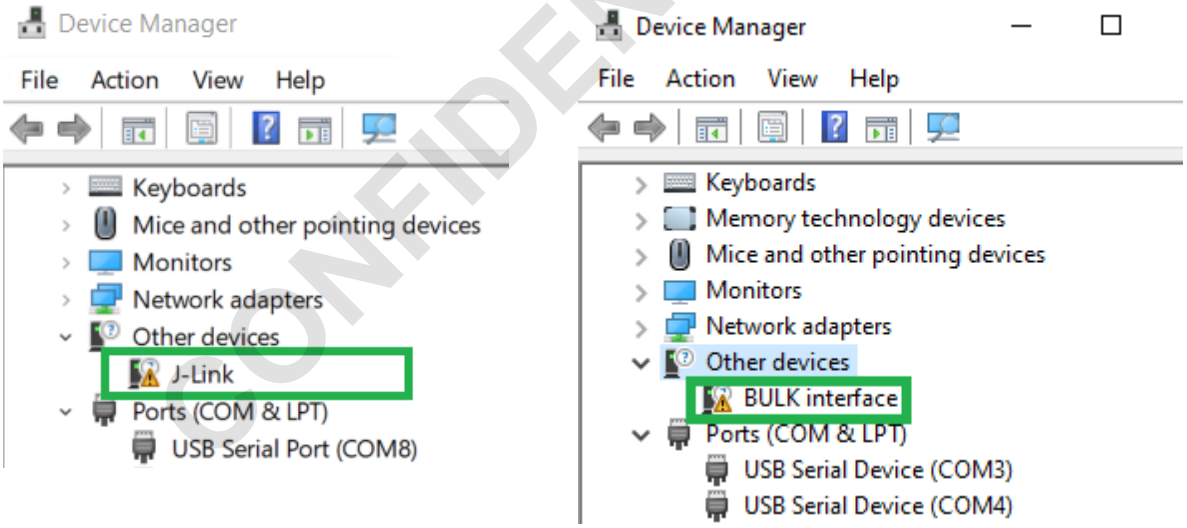
Enter `devmgmt.msc` then click the `OK` button to verify whether the J-Link OB interface or BULK interface exists or not.

If there is a J-Link or BULK interface listed in the device manager under `Other devices`, please click the `Install RDI Interface` shortcut in `Start Menu/Atmosic` folder.

***Note**: The COM port numbers are assigned by the Windows system.*

***Note**: If the device manager shows Install Completed, move to the next section.*

Open Device Manager before installing the RDI Interface driver.



If J-Link driver already installed

No previously installed J-Link driver

Figure 15 - USB COM Port with ATM33/e EVK

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Please click the `Install` button if the system pop-up the following message.

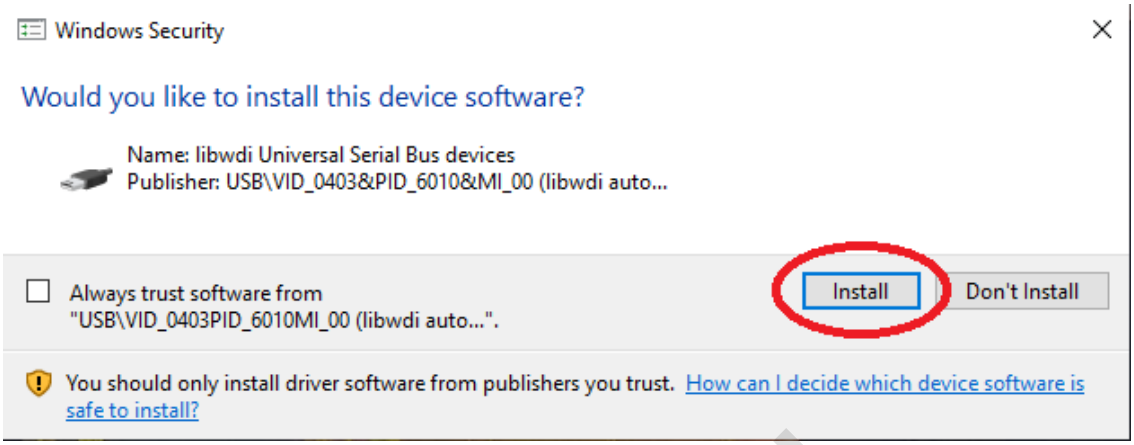


Figure 16 - Install RDI USB Driver with ATM33/e EVK

After installing Atmosic RDI USB driver, verify the successful installation and confirm that the BULK interface shows up under Universal Serial Bus devices in Device Manager. Moreover, the second serial COM port shown in these examples, COM4 is the debug console output that displays the debug messages through PuTTY.

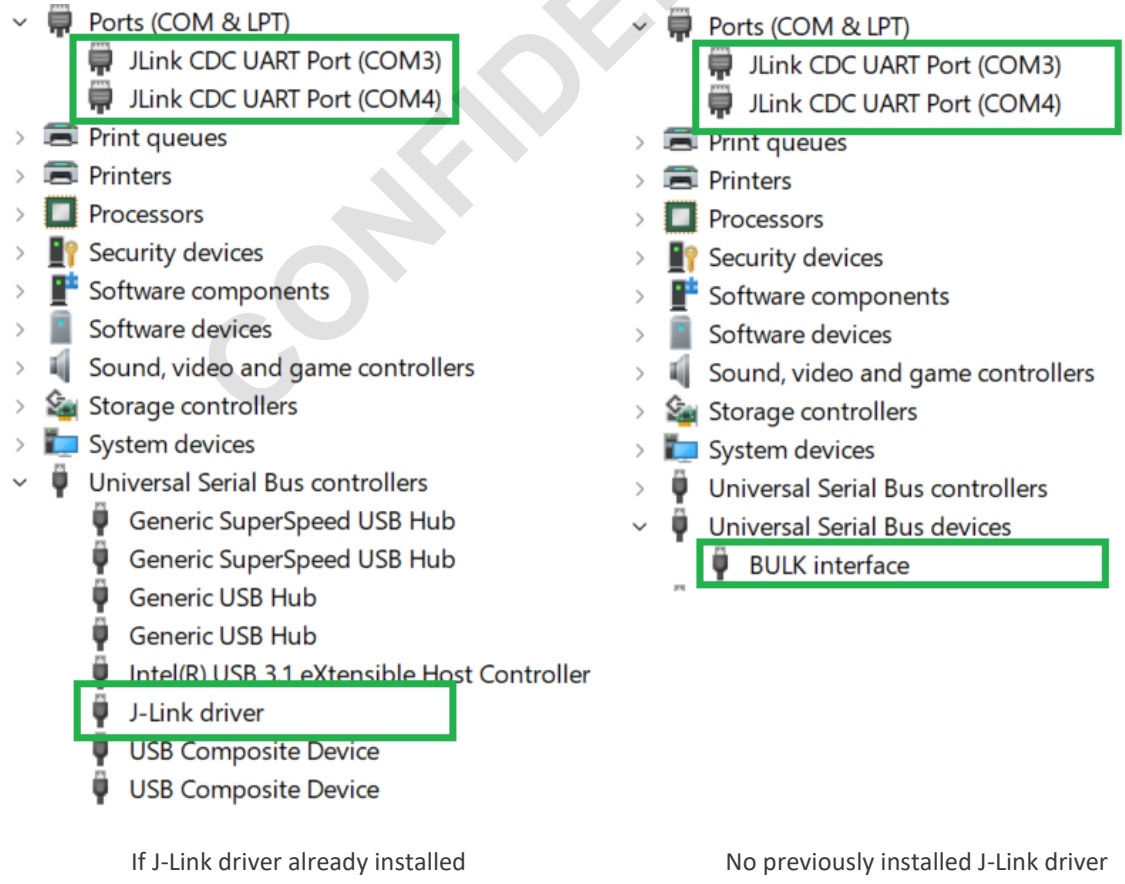


Figure 17 - RDI USB in Device Manager with ATM33/e EVK

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3.3.3 Remove RDI Interface

If you want to remove the RDI driver or switch to JLink, please click the Remove RDI Interface shortcut in Start Menu/Atmosic folder as shown in Figure 18.

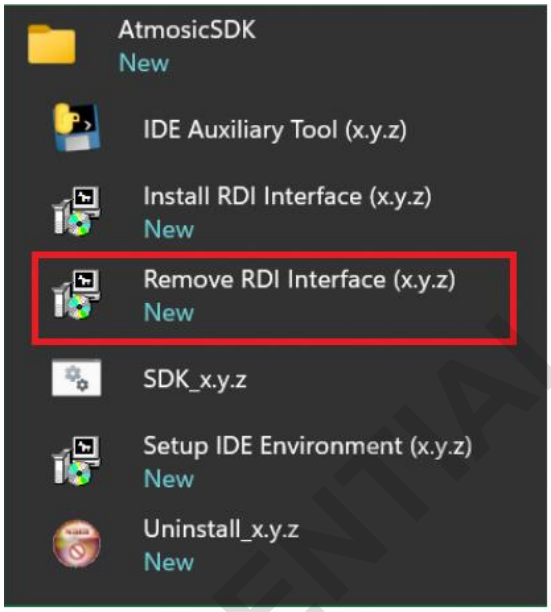


Figure 18 - Remove RDI Interface with ATM33/e EVK

3.4 Connect To Debug Interface

The Atmosic EVK provides the log output through a debug interface (COM port). The developer could use utility to capture (i.e. PuTTY, Tera Term ...etc.) the log. This section describes how to capture logs through PuTTY.

When opening the PuTTY tool, click the

1) `Serial` radio button then input the COM port information in
2) `Serial line` item (i.e. COM11),

and input 115200 in

3) `Speed` item.

Subsequently, click the

4) `Open` button to open the COM port and wait for debug message output.

The debug console will only begin to display when an example is running; otherwise the console will remain empty as shown in Figure 19.

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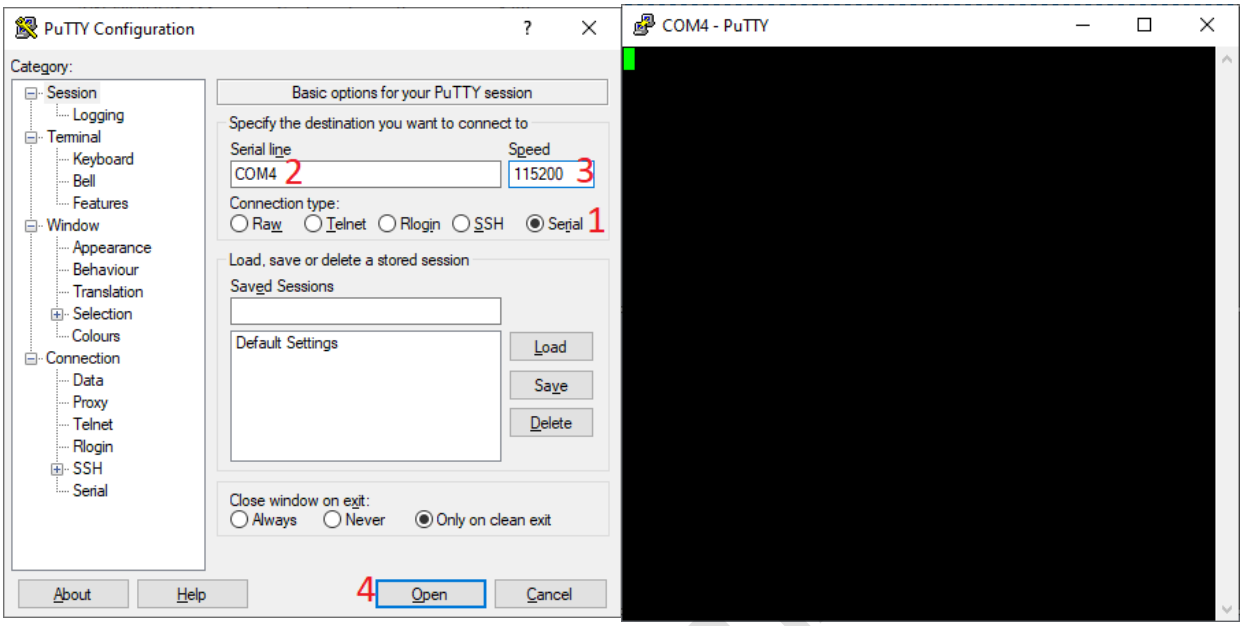


Figure 19 - Debug Interface Using PuTTY

3.5 Launch MSYS2 shell

Invoke the MSYS2 shell from the Windows menu, Start Menu/AtmosicSDK/SDK_<version_number>. Figure 20 shows the Start Menu.

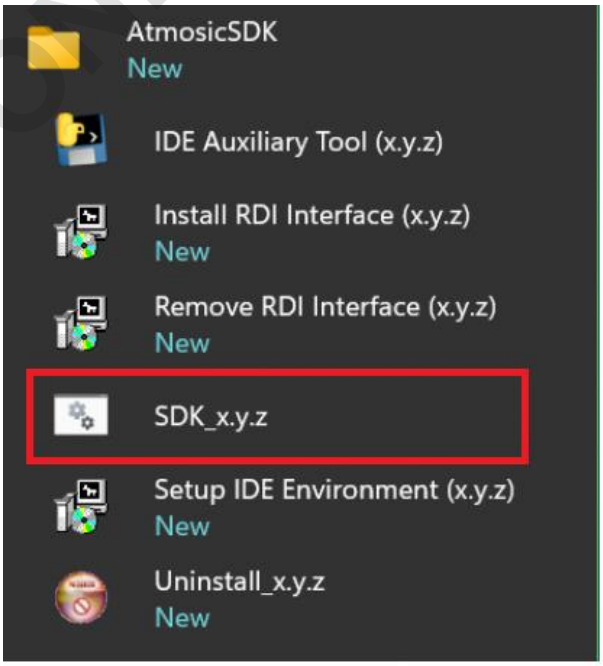


Figure 20 - Start Menu

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The MSYS2 shell will be invoked and located at the AtmosicSDK working directory as shown in Figure 21.

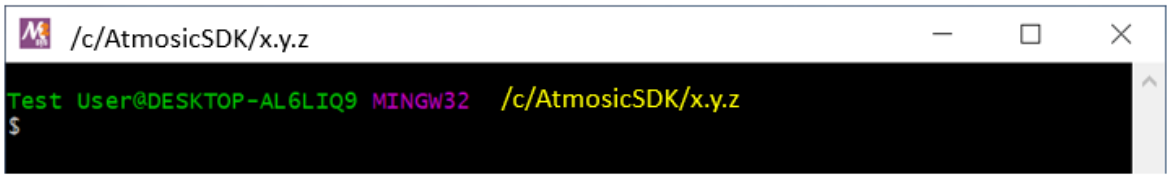


Figure 21 - Atmosic SDK Path

3.6 Atmosic SDK Platform Working Directory

SDK 5.2.0 and later supports ATM2/ATM3 and ATM33/e devices. The difference between ATM2 and ATM3 device versions is captured by the second digit in the -xxx extension of the product code for the device.

In order to build the correct application software for any ATM2, ATM3, or ATM33/e devices, it is necessary to identify the device version and use the appropriate directory in the Atmosic SDK for the software build.

ATM2 and ATM3 EVB have a sticker that identifies the part number of the Atmosic device on the board. For example, the Atmosic EVB is labeled with ATM3221-010 as shown in Figure 22. It is assembled with an ATM32xx-x1x chip revision and to compile for this revision, the platform path is platform/atm3/ATM32xx-x1x. Refer to the Atmosic Chip Revisions Application Note for information regarding ATM2/ATM3 chip revisions.

The ATM3 EVB has printed the part number of the Atmosic device on the board, please refer to Figure 22.

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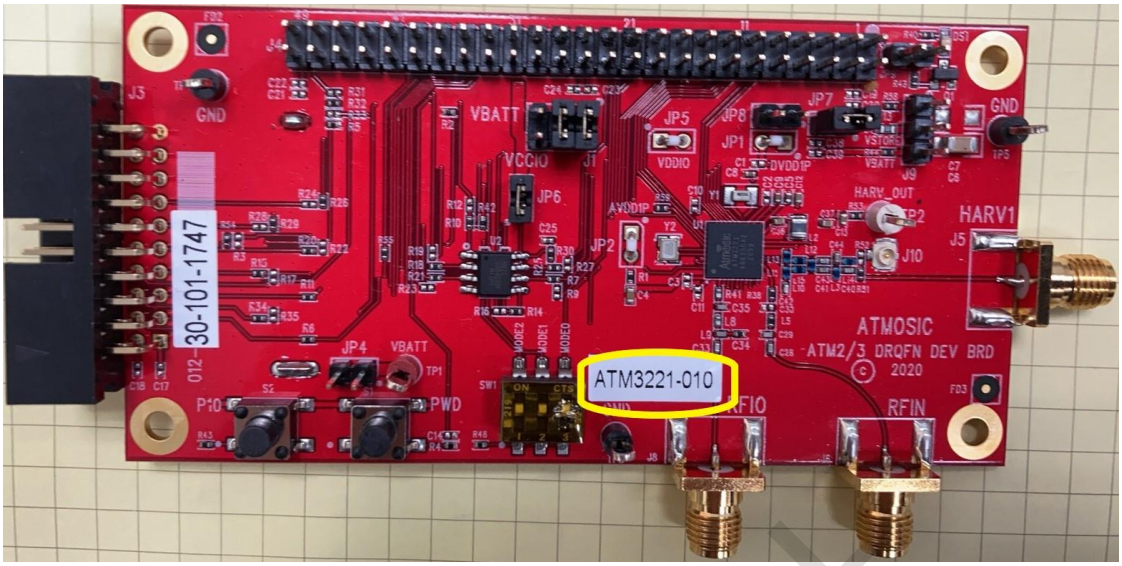


Figure 22 - Atmosic Product Code on the EVB

Following is a list of all Atmosic SDK platform working directories, please use the appropriate directory in the Atmosic SDK for the software build.

- platform/atm2/ATM22xx-x0x
- platform/atm2/ATM22xx-x1x
- platform/atm3/ATM32xx-x0x
- platform/atm3/ATM32xx-x1x
- platform/atm33/ATM33xx-5

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4. Building Beacon Application Using Atmosic SDK

4.1 ATM2 and ATM3 Device

At MSYS2 shell, go to the BLE_adv example folder to compile the project through the following command.

4.1.1 Board Configuration

The examples that depend on the pinmux driver directly or indirectly (via the gpio, i2c, spi, keyboard, pdm, gadc drivers) will need to specify the board for which the example is being built.

For ATM22xx and ATM32xx devices, the EVB with specific SoC can be specified as: m2201, m2202, m2221, m2231, m2251, m3201, m3202, m3221, and m3231.

4.1.2 Beacon

Beacon configuration: EddyStone, Non Connectable, Scannable, 100 ms Interval.

```
$ cd platform/atm2/ATM22xx-x1x/examples/BLE_adv
$ make BOARD=<chip>
```

If everything is correct, the `make` should execute without error as shown below:

```
user@labtop MINGW32 /c/AtmosicSDK/x.y.z
$ cd platform/atm2/ATM22xx-x1x/examples/BLE_adv

user@labtop MINGW32 /c/AtmosicSDK/x.y.z/platform/atm2/ATM22xx-x1x/examples/BLE_adv
$ make BOARD=m2221
```

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```
user@labtop MINGW32 /c/AtmosicSDK/x.y.z/platform/atm2/ATM22xx-x1x/examples/BLE_adv
$ ls
BLE_adv.asm       atm_gap.o                  pinmux.o
BLE_adv.bin       atm_gap_param.d            pmu.d
BLE_adv.c         atm_gap_param.o            pmu.o
BLE_adv.d         atm_pm.d                   porting_ble.d
BLE_adv.elf       atm_pm.o                   porting_ble.o
BLE_adv.h         ble_gap.d                  reference_beacons.mk
BLE_adv.map       ble_gap.o                  reset.d
BLE_adv.o         ble_module.d               reset.o
README            ble_module.o               rf.d
ROM_errata_10.d   ble_task.d                 rf.o
ROM_errata_10.o   ble_task.o                 rom.elf
ROM_errata_14.d   brwnout.d                  rom.o
ROM_errata_14.o   brwnout.o                  sw_timer.d
ROM_errata_16.d   config_adv_params.h        sw_timer.o
ROM_errata_16.o   ext_flash.d                swd_dbg.d
arm_traceback.d   ext_flash.o                swd_dbg.o
arm_traceback.o   flash_nvds.bin             tag_data
atm_adv.d         hardfault_handler_armv6m.d timer.d
atm_adv.o         hardfault_handler_armv6m.o timer.o
atm_adv_param.d   heap.d                     user_debug.d
atm_adv_param.o   heap.o                     user_debug.o
atm_asm.d         hw_cfg.d                   user_init.d
atm_asm.o         hw_cfg.o                   user_init.o
atm_ble.d         interrupt.d                user_startup_CMSDK_CM0.o
atm_ble.o         interrupt.o                watchdog.d
atm_debug.d       makefile                   watchdog.o
atm_debug.o       partition_info.map
atm_gap.d         pinmux.d
```

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At MSYS2 shell, type `make BOARD=<chip> run_all` to build and load firmware into Atmosic EVK.

Shell and debug terminals after the `make BOARD=<chip> run_all` command is successfully executed.

```
@00000036 SDK Version: x.y.z
@0000008c APP Version: 0.0.0.9
6x6 EXT_FLASH: 4e 56 44 53 06 06 27 00 00 00 00 01 02 00 00 00 ...
@000001de boot_status = 1000000
@00000239 Cold boot
@00000274 [   BLE_adv][D]: user_main() done
@00000344 [   BLE_adv][D]: ble_adv_init: NVDS tag for adv timeout param not found. Using default
@000004f9 [   atm_adv][D]: Adv dur 0(unit:10ms) max_adv_evt 0 (timeout 0ms)
@00000608 [   BLE_adv][D]: adv_state = 2
@000006b6 [   BLE_adv][D]: adv_state = 4
@00000758 [   BLE_adv][D]: adv_state = 6
@000007e8 Optimize HW Scan: Active=1
@0000086a [   atm_adv][D]: Adv0: ON
@000008eb [   BLE_adv][D]: adv_state = 9
```

Detailed documentation for Beacon and other examples in SDK is available in platform/atm2/ATM2xx-x1x/examples/<example_name>/README file.

4.2 ATM33/e Device

At MSYS2 shell, go to the BLE_adv example folder to compile the project using the following command.

4.2.1 Board Configuration

The examples that depend on the pinmux driver directly or indirectly (via the gpio, i2c, spi, keyboard, pdm, gadc drivers) will need to specify the board for which the example is being built.

For ATM33xx device, the EVB with specific SoC boards can be specified as:

- ATMEVK_3330_QN
- ATMEVK_3330e_QN
- ATMEVK_3325_QK

4.2.2 Beacon

Beacon configuration: EddyStone, Non Connectable, Scannable, 100 ms Interval.
To build and load firmware into the Atmosic EVK, follow below:

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```
$ cd platform/atm33/ATM33xx-5/examples/BLE_adv/
$ make BOARD=<chip> run_all
```

***Note**: Reminder to install COIN battery when VBATT = COIN OR configure EVB to run on VBATT = 3 V. Refer to the EVK User Guides for specific EVB for correct jumper settings.*

5. Building Energy Harvesting Beacon

5.1 ATM3 Devices

At MSYS2 shell, go to the BLE_harv_adv example folder to compile the project through the following command.

Beacon configuration: EddyStone, Non Connectable, Scannable, 1s Interval.

```
$ cd platform/atm3/ATM32xx-x1x/examples/BLE_harv_adv/
$ make BOARD=<chip> DEBUG= run_all
```

5.2 ATM33e Devices

At MSYS2 shell, go to the BLE_harv_adv example folder to compile the project using the following command.

Beacon configuration: EddyStone, Non Connectable, Scannable, 1s Interval.

```
$ cd platform/atm33/ATM33xx-5/examples/BLE_harv_adv/
$ make BOARD=<chip> DEBUG=  run_all
```

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6. Uninstall Atmosic SDK

Follow the steps to uninstall Atmosic SDK. Also see Figure 23.

1) Click the `Start Menu` and `Settings`.
2) In the `Settings` page, click the `Apps` item.
3) Find the `Atmosic SDK (<version_number>)` item in the `Apps & features` page.
4) Click the `Uninstall` button to start the Atmosic SDK uninstallation process.
5) The Atmosic RDI Interface will be removed during the uninstallation process. Please remove the Atmosic EVB before reinstalling the Atmosic SDK again.

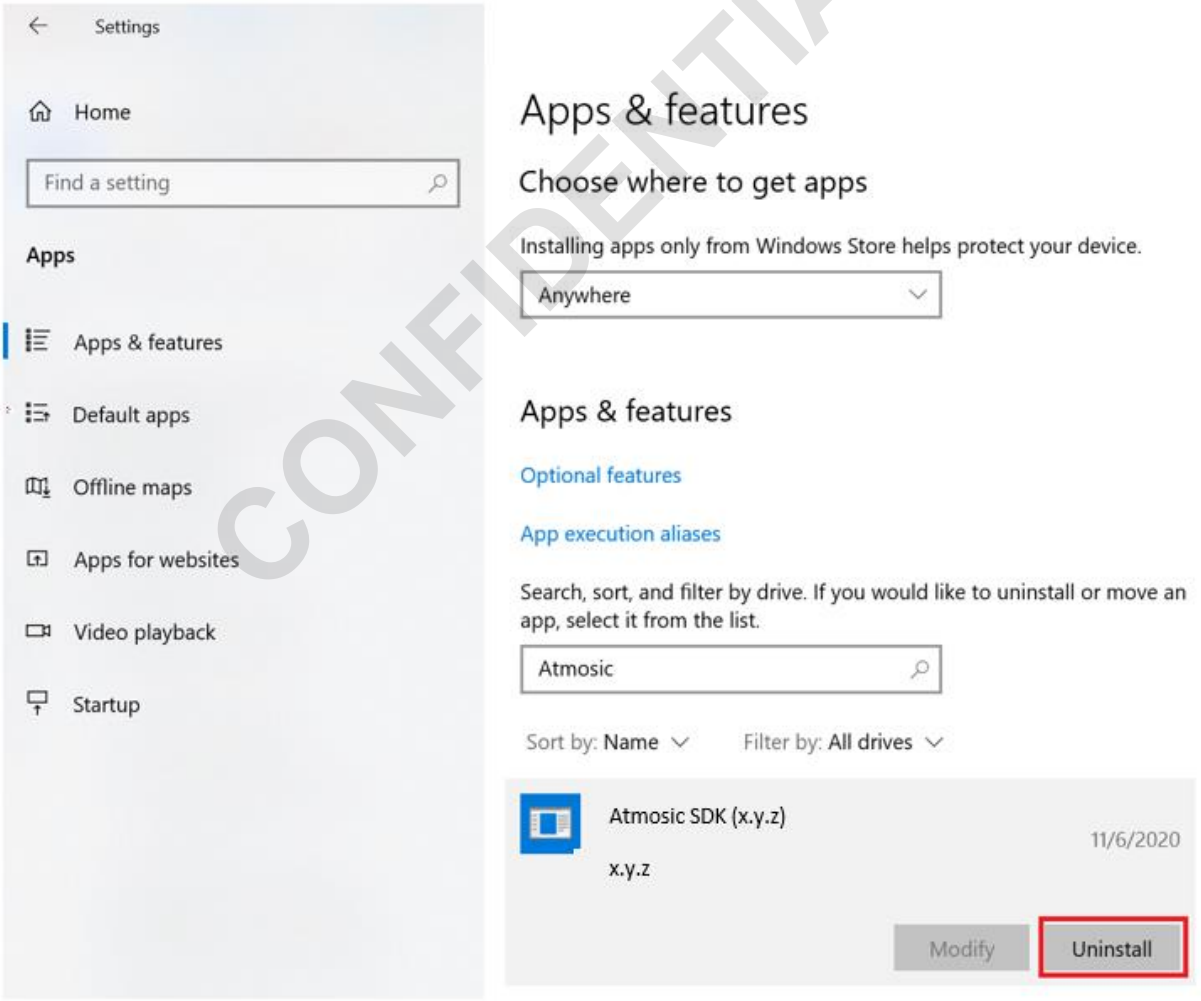


Figure 23 - Uninstall SDK

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Reference Documents

Title	Document Number
ATM2/ATM3 EVK Power Consumption Evaluation User Guide	ATM2_ATM3-UGPCE
EVK User's Guide for ATMx301/ATMx202	ATMx201-UG
EVK User's Guide for ATMx221	ATMx221-UG
ATM33/e Series EVK Power Consumption Evaluation User Guide	ATM33_e-UGPCE
ATM33/e Series Evaluation Kit User Guide	ATM33_e-URGEVK
ATM33e Series Energy Harvesting EVK Quick Start Guide	ATM33e-QSGEHV
IAR Workbench User Guide	ATM-UGIAR
Keil MDK Configuration Guide	ATM-CGKMDK
SEGGER Embedded Studio User Guide	ATM-UGSEG
IDE Auxiliary Flash Programming Tool User Guide	ATM-UGIDEFPT

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Revision History

Date	Version	Description
November 9, 2022	0.61	Added Remove RDI Interface section. Updated ATM33/e EVK, Building Beacon Application Using Atmosic SDK and Building Energy Harvesting Beacon sections.
May 13, 2022	0.60	Initial version created for SDK 5.1.0 release.

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